

# Lecture 4: Fundamentals of Antenna Theory

Chin-Fei Lee

`cflee@asiaa.sinica.edu.tw`

# Electromagnetic Potentials

Direct solns of Maxwell's eqs. are very difficult for all but the simplest configurations. Introduce new functions of  $\mathbf{E}$  and  $\mathbf{B}$  that can be determined from the imposed **current density  $\mathbf{J}$  and the charge density  $\rho$** . From these both  $\mathbf{E}$  and  $\mathbf{B}$  can then be determined in a straightforward way. Assuming that  $\epsilon$ ,  $\mu$  and  $\sigma$  are constant, then

$$(1) \quad \nabla \cdot \mathbf{B} = 0 \quad \Rightarrow \quad \mathbf{B} = \nabla \times \mathbf{A} \quad (\because \nabla \cdot \nabla \times \mathbf{A} = 0)$$

$$(2) \quad \nabla \times \mathbf{E} = -\frac{1}{c} \dot{\mathbf{B}} \quad \Rightarrow \quad \mathbf{E} = -\nabla \Phi - \frac{1}{c} \dot{\mathbf{A}} \quad (\because \nabla \times \nabla \Phi = 0)$$

$$(3) \quad \nabla \times \mathbf{H} = \frac{4\pi}{c} \mathbf{J} + \frac{1}{c} \dot{\mathbf{D}} \quad \Rightarrow \quad \nabla^2 \mathbf{A} - \frac{\epsilon\mu}{c^2} \ddot{\mathbf{A}} - \nabla(\nabla \cdot \mathbf{A} + \frac{\epsilon\mu}{c} \dot{\Phi}) = -\frac{4\pi}{c} \mu \mathbf{J}$$

$$(4) \quad \nabla \cdot \mathbf{D} = 4\pi\rho \quad \Rightarrow \quad \nabla^2 \Phi - \frac{\epsilon\mu}{c^2} \ddot{\Phi} - \frac{1}{c} \frac{\partial}{\partial t}(\nabla \cdot \mathbf{A} + \frac{\epsilon\mu}{c} \dot{\Phi}) = -\frac{4\pi}{\epsilon} \rho$$

where  $\mathbf{A}$  and  $\Phi$  are vector and scalar electrodynamic potentials, respectively. Here  $\nabla \times (\nabla \times \mathbf{A}) = \nabla^2 \mathbf{A} - \nabla(\nabla \cdot \mathbf{A})$ . Simplifying by using the **Lorentz gauge**:

$$\boxed{\nabla \cdot \mathbf{A} + \frac{\epsilon\mu}{c} \dot{\Phi} = 0}$$

# Lorentz Gauge: A way to simplify thing!

Note that in Lorentz gauge, the vector and scalar potentials are not unique!

(1)  $\because \nabla \times \nabla \Lambda = 0$ , adding a vector  $\nabla \Lambda$  to  $\mathbf{A}$  doesn't change the resulting  $\mathbf{B}$ .

$$\hat{\mathbf{A}} = \mathbf{A} + \nabla \Lambda$$

where  $\Lambda$  is a scalar gauge function that can be derived.

(2) Adding a corresponding term to keep  $\mathbf{E}$  unchanged. ( $\hat{\dot{\mathbf{A}}} = \dot{\mathbf{A}} + \nabla \dot{\Lambda}$ )

$$\hat{\Phi} = \Phi - \frac{1}{c} \dot{\Lambda}$$

Substituting above two eqs. into the Lorentz gauge, we find that the new set of  $\mathbf{A}$  and  $\Phi$  are also in Lorentz gauge as long as  $\Lambda$  fulfills the following condition:

$$\nabla^2 \Lambda - \frac{\epsilon \mu}{c^2} \ddot{\Lambda} = 0$$

Anyway, with the Lorentz gauge, we end up with

$$(5) \quad \nabla^2 \mathbf{A} - \frac{\epsilon \mu}{c^2} \ddot{\mathbf{A}} = -\frac{4\pi}{c} \mu \mathbf{J}$$

$$(6) \quad \nabla^2 \Phi - \frac{\epsilon \mu}{c^2} \ddot{\Phi} = -\frac{4\pi}{\epsilon} \rho$$

$\Rightarrow$  decoupled inhomogeneous wave eqs.

# How to solve? 4 steps to the soln!

The inhomogeneous wave eqs have the form

$$\nabla^2 \psi - \frac{1}{v^2} \ddot{\psi} = -f(\mathbf{x}, t)$$

with

$$v = \frac{c}{\sqrt{\epsilon\mu}}$$

(1) Removing  $t$  dependence by taking the inverse Fourier Transform, i.e., decomposing the wave into independent components with different  $\omega$ . With the Fourier Transforms:

$$\begin{aligned}\psi(\mathbf{x}, t) &= \int_{-\infty}^{\infty} \Psi(\mathbf{x}, \omega) e^{i\omega t} d\omega \\ f(\mathbf{x}, t) &= \int_{-\infty}^{\infty} F(\mathbf{x}, \omega) e^{i\omega t} d\omega\end{aligned}$$

we end up with a time-independent Helmholtz wave eq.

$$(\nabla^2 + k^2) \Psi(\mathbf{x}, \omega) = -F(\mathbf{x}, \omega)$$

with  $k = \omega/v$ .

# Green's Function

(2) [Solving Green's Function](#). Assuming  $F$  (i.e., the charge distribution) can be expanded into a linear combination of delta functions (i.e., point charge),

$$F(\mathbf{x}, \omega) = \int F(\mathbf{x}', \omega) \delta(\mathbf{x} - \mathbf{x}') d^3 x'$$

then  $\Psi$  is formed by the same kind of linear superposition (or combination) as  $F$ ,

$$\Psi(\mathbf{x}, \omega) = \int F(\mathbf{x}', \omega) G(\mathbf{x} - \mathbf{x}') d^3 x'$$

with the [Green's function being the soln of the delta function](#):

$$(\nabla^2 + k^2)G(\mathbf{x}, \mathbf{x}') = -\delta(\mathbf{x} - \mathbf{x}')$$

In the electrostatic example, this corresponds to building up the total potential field from the fields of individual point charges, each situated at the position  $\mathbf{x}'$ .

# Green Functions

(3) Take the inverse Fourier Transform of the Green function (restoring  $t$  dependence).

$$g(\mathbf{x}, t, \mathbf{x}', t') = \frac{1}{2\pi} \int G(\mathbf{x}, \mathbf{x}') e^{i\omega t} d\omega$$

Here  $g$  can be considered as the potential soln for each point charge located at  $\mathbf{x} = \mathbf{x}'$  and  $t = t'$ .

(4) Find the final soln for the wave eq. by integrating over time and space:

$$\psi(\mathbf{x}, t) = \int \int f(\mathbf{x}', t') g(\mathbf{x}, t, \mathbf{x}', t') d^3x' dt'$$

# Spherical wave from a point charge

The Green's function method can be applied to the case of **spherical potential waves** from a point charge. In spherical coordinate  $(r, \vartheta, \varphi)$  and charge at  $r = 0$ , we have

$$\frac{1}{r} \frac{d^2}{dr^2} (rG) + k^2 G = -\delta(r)$$

For  $r \neq 0$ , the soln is

$$G = \frac{1}{4\pi r} e^{\pm ikr}$$

Take the inverse Fourier transform,

$$g(\mathbf{x}, \mathbf{x}', t) = \frac{1}{4\pi |\mathbf{x} - \mathbf{x}'|} \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i(\omega t \pm k|\mathbf{x} - \mathbf{x}'|)} d\omega$$

Introducing

$$t' = t \mp \frac{k}{\omega} |\mathbf{x} - \mathbf{x}'| = t \mp \frac{|\mathbf{x} - \mathbf{x}'|}{v} \Rightarrow t' = t - \frac{|\mathbf{x} - \mathbf{x}'|}{v}$$

For proper causal relation, only take the upper sign, which means **retarded time**, so

$$g(\mathbf{x}, \mathbf{x}', t, t') = \frac{\delta(t' + \frac{|\mathbf{x} - \mathbf{x}'|}{v} - t)}{4\pi |\mathbf{x} - \mathbf{x}'|}$$

Because of finite wave propagation speed, **the field of the wave at  $(\mathbf{x}, t)$  depends on charge at  $(\mathbf{x}', t')$  in the past.**

# Spherical wave from a group of charge

In the absence of boundaries, the potential soln can be found by

$$\psi(\mathbf{x}, t) = \frac{1}{4\pi} \int \int \frac{f(\mathbf{x}', t') \delta(t' + \frac{|\mathbf{x} - \mathbf{x}'|}{v} - t)}{|\mathbf{x} - \mathbf{x}'|} d^3x' dt'$$

Integrating over  $t'$ , we end up with

$$\psi(\mathbf{x}, t) = \frac{1}{4\pi} \int \frac{f(\mathbf{x}', t - \frac{|\mathbf{x} - \mathbf{x}'|}{v})}{|\mathbf{x} - \mathbf{x}'|} d^3x'$$

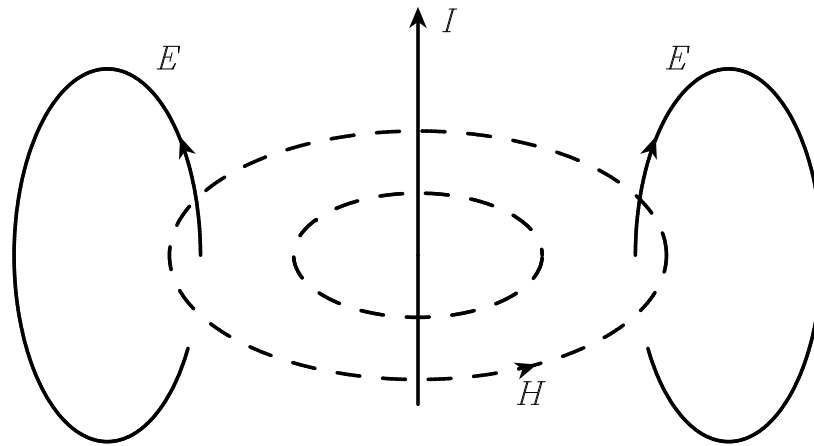
The electrodynamics potentials for the Maxwell's eq. are then given

$$A(\mathbf{x}, t) = \frac{\mu}{c} \int \int \int \frac{\mathbf{J}(\mathbf{x}', t - \frac{|\mathbf{x} - \mathbf{x}'|}{v})}{|\mathbf{x} - \mathbf{x}'|} d^3x' \quad (\because f = \frac{4\pi}{c} \mu \mathbf{J})$$

$$\Phi(\mathbf{x}, t) = \frac{1}{\epsilon} \int \int \int \frac{\rho(\mathbf{x}', t - \frac{|\mathbf{x} - \mathbf{x}'|}{v})}{|\mathbf{x} - \mathbf{x}'|} d^3x' \quad (\because f = \frac{4\pi}{\epsilon} \rho)$$

Note that a soln specifying  $\mathbf{A}$  and  $\Phi$  (or  $\mathbf{E}$  and  $\mathbf{B}$ ) will result in currents and charges. These must be self-consistent, and this is a coupled problem and will not be discussed.

# EM wave induced by a current



A change of current can produce a EM wave propagating away. A steady current can induce a  $\mathbf{H}$  field around it. As the current changes, the  $\mathbf{H}$  field changes, inducing a  $\mathbf{E}$  field and thus producing a wave propagating away.

# Hertz Dipole as a simplest example

Let's calculate a dipole field from an antenna with a length  $\Delta l$  and a cross section  $q$ . In this case, the currents are confined to a finite region, the antenna proper, and we can postulate a given distribution for  $\mathbf{J}$ . Moreover, any reactions that the field has on these currents are assumed to be negligible.

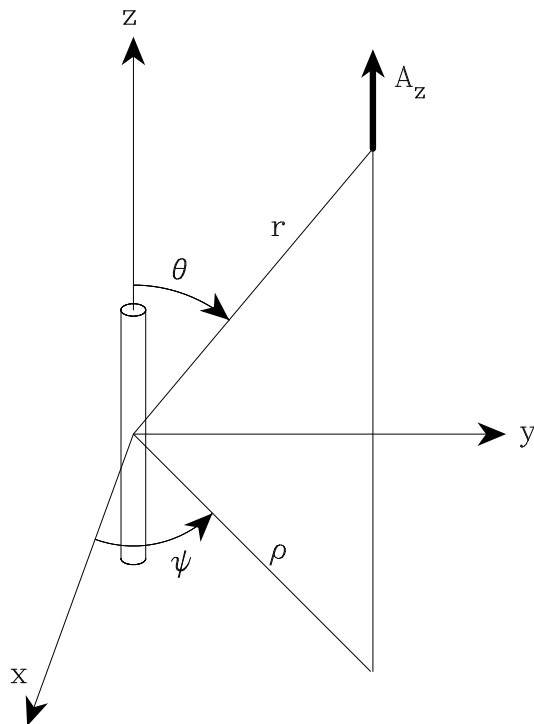
Setting up a simple wavelike current  $I$  in this dipole, then

$$\mathbf{J} = \frac{I}{q} e^{-i\omega t} \hat{z}$$

Derive  $\mathbf{A}$  then  $\mathbf{B}$  first with  $\mathbf{J}$ , and then  $\mathbf{E}$  with Maxwell's eq. The dipole extends from  $z = -\Delta l/2$  to  $+\Delta l/2$  and  $A_z$  is

$$\begin{aligned} A_z &= \frac{\mu}{c} \int_{-\Delta l/2}^{+\Delta l/2} \frac{I/q}{r} \exp\left[-i\omega\left(t - \frac{r}{v}\right)\right] q dl \\ &= \frac{\mu}{c} \frac{I \Delta l}{r} \exp[-i(\omega t - kr)] \end{aligned}$$

$A_z$  is the only nonzero component and depends only on  $r$ .



# Hertz Dipole

Introduce a cylindrical coordinate  $(\rho, \phi, z)$  since  $A_z$  is the only nonzero component of  $\mathbf{A}$  and depends only on  $r^2 = \rho^2 + z^2$ , so that only  $B_\phi$  is not zero and given by

$$B_\phi = (\nabla \times \mathbf{A})_\phi = \frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} = -\frac{\partial A_z}{\partial \rho} = -\frac{\partial A_z}{\partial r} \frac{\partial r}{\partial \rho}$$

Since

$$\frac{\partial r}{\partial \rho} = \frac{\rho}{r} = \sin \theta \quad (\because r^2 = \rho^2 + z^2, \quad \rho = r \sin \theta)$$

$$\frac{\partial A_z}{\partial r} = i \frac{\mu}{c} \frac{k I \Delta l}{r} \left[ 1 - \frac{1}{i k r} \right] \exp[-i(\omega t - k r)]$$

We end up with (book uses modified c.g.s  $J \Rightarrow \frac{c}{4\pi} J$ , so  $\frac{k}{c} \Rightarrow \frac{c}{4\pi} \frac{k}{c} = \frac{1}{2\lambda}$  with  $k = \frac{2\pi}{\lambda}$ )

$$H_\phi = \frac{1}{\mu} B_\phi = -i \frac{k I \Delta l \sin \theta}{c r} \left[ 1 - \frac{1}{i k r} \right] \exp[-i(\omega t - k r)]$$

Thus a change in current produces a magnetic field propagating away with  $v = \omega/k$ .

# Hertz Dipole

The changes in  $\mathbf{B}$  causes a changes in  $\mathbf{E}$  according to the Maxwell's eq. Outside the dipole,  $\sigma = 0$  and thus  $\mathbf{J} = 0$ . For a harmonic wave,  $\dot{\mathbf{D}} = -i\omega\mathbf{D} = -i\omega\epsilon\mathbf{E}$ , thus with the Ampere-Maxwell law

$$\mathbf{E} = \frac{ic}{\omega\epsilon}(\nabla \times \mathbf{H}) = \frac{i}{k} \sqrt{\frac{\mu}{\epsilon}}(\nabla \times \mathbf{H})$$

using  $v = c/\sqrt{\epsilon\mu}$ . Returning to the spherical coordinate system  $(r, \theta, \phi)$ , we have

$$E_{\theta} = \frac{i}{k} \sqrt{\frac{\mu}{\epsilon}}(\nabla \times \mathbf{H})_{\theta} = \frac{i}{k} \sqrt{\frac{\mu}{\epsilon}} \left[ -\frac{1}{r} \frac{\partial(rH_{\phi})}{\partial r} \right]$$

$$E_r = \frac{i}{k} \sqrt{\frac{\mu}{\epsilon}}(\nabla \times \mathbf{H})_r = \frac{i}{k} \sqrt{\frac{\mu}{\epsilon}} \left[ \frac{1}{r \sin \theta} \frac{\partial(\sin \theta H_{\phi})}{\partial \theta} \right]$$

$$E_{\phi} = 0$$

# Hertz Dipole: Full soln

Thus,

$$H_{\phi} = -i \frac{kI\Delta l \sin \theta}{cr} \left[ 1 - \frac{1}{ikr} \right] \exp[-i(\omega t - kr)]$$

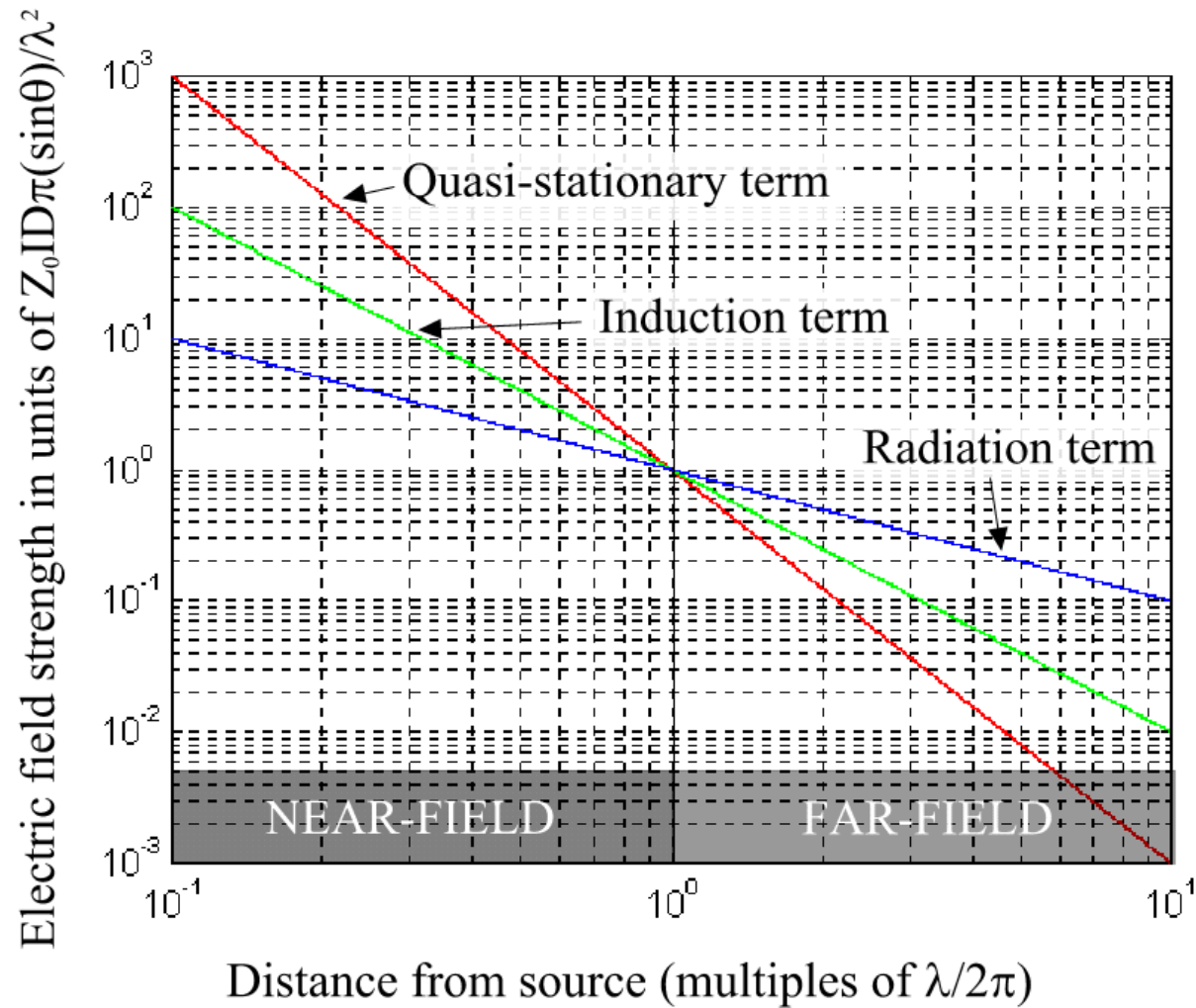
$$E_{\theta} = -i \sqrt{\frac{\mu}{\epsilon}} \frac{kI\Delta l \sin \theta}{cr} \left[ 1 - \frac{1}{ikr} + \frac{1}{(ikr)^2} \right] \exp[-i(\omega t - kr)]$$

$$E_r = i \sqrt{\frac{\mu}{\epsilon}} \frac{kI\Delta l \cos \theta}{cr} \left[ \frac{1}{ikr} - \frac{1}{(ikr)^2} \right] \exp[-i(\omega t - kr)]$$

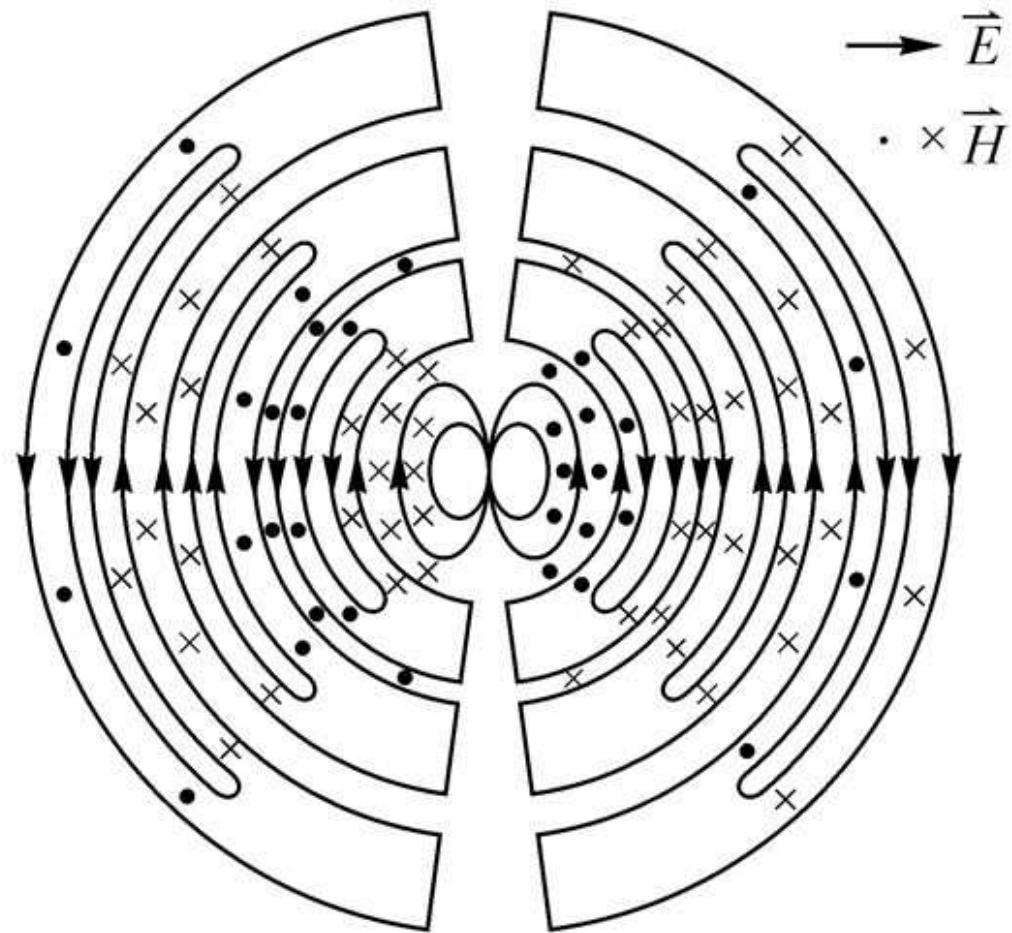
Since  $\mathbf{E} \cdot \mathbf{H} = 0$ ,  $\mathbf{E}$  and  $\mathbf{H}$  of a radiating dipole are **perpendicular** everywhere.

- Terms with  $1/r^2$ : induction field of a slowly oscillating electric dipole.
- Terms with  $1/r^3$ :  $\mathbf{E}$  field of the static dipole.
- Terms with  $1/r$ : radiation field (dominate the far field)

# Hertz Dipole

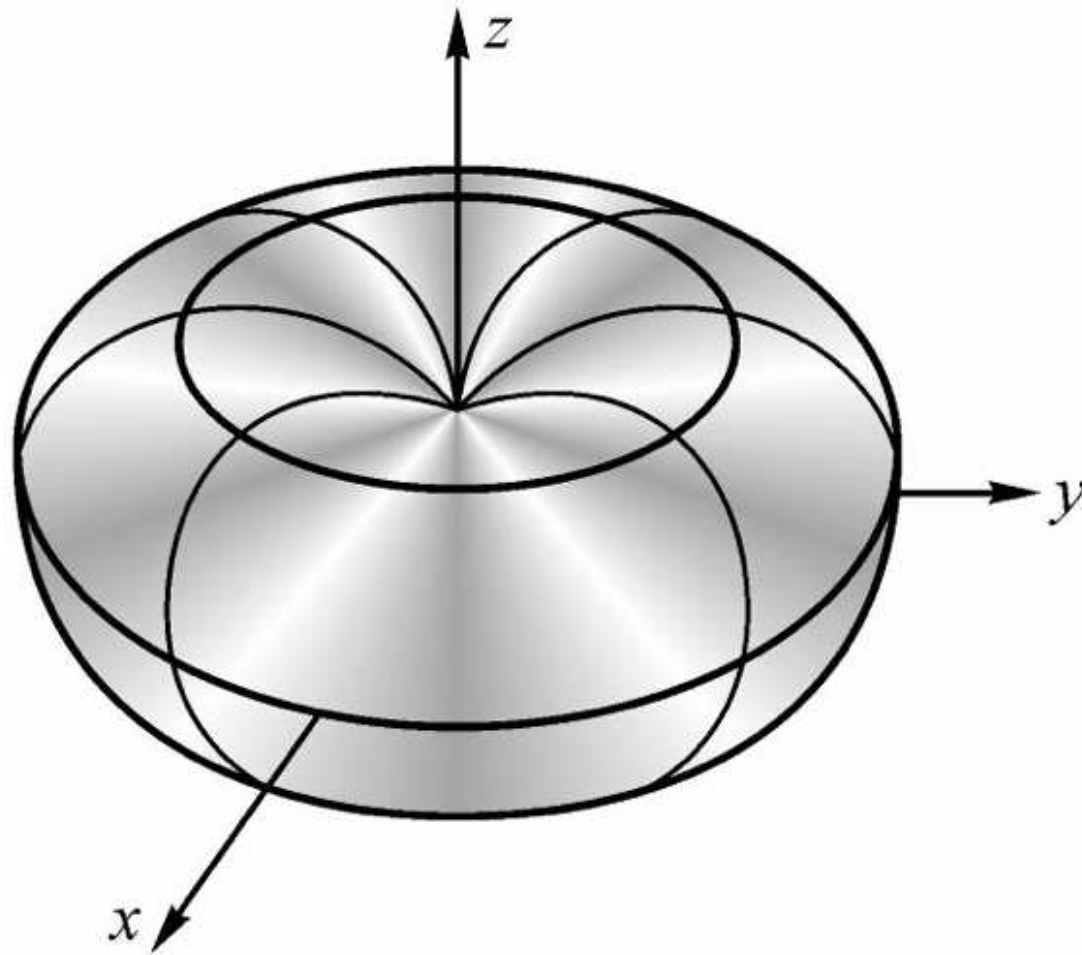


# Hertz Dipole: Field Cross-Section



Note that  $\vec{E} \parallel \vec{J}$  in the equator and  $\vec{E} \perp \vec{B}$ . The wave propagates in the direction of  $\vec{E} \times \vec{H}$ .

# Hertz Dipole: 3D Field Pattern of a dipole



The field pattern is torus-shaped perpendicular to the dipole axis, ignoring the oscillation.

# Hertz Dipole: Radiation field

Thus,

$$H_{\phi} = -i \frac{kI\Delta l}{c} \frac{\sin \theta}{r} \exp[-i(\omega t - kr)]$$

$$E_{\theta} = -i \sqrt{\frac{\mu}{\epsilon}} \frac{kI\Delta l \sin \theta}{cr} \exp[-i(\omega t - kr)]$$

⇒ **spherically expanding wave**. Just as in the case of plane EM waves,

$$\frac{|\mathbf{E}|}{|\mathbf{H}|} = \sqrt{\frac{\mu}{\epsilon}}$$

And the Poynting vector is directed radially outward, its time average is

$$|\langle \mathbf{S} \rangle| = \frac{c}{4\pi} |\operatorname{Re}(\mathbf{E} \times \mathbf{H}^*)| = \frac{c}{4\pi} \sqrt{\frac{\mu}{\epsilon}} \mathbf{H}^2 = \frac{c}{4\pi} \sqrt{\frac{\epsilon}{\mu}} \mathbf{E}^2$$

# Hertz Dipole

Thus, the total radiated power, with  $\epsilon \sim 1$  and  $\mu \sim 1$ , is

$$\begin{aligned} P &= \int_0^{2\pi} \int_0^\pi |\langle \mathbf{S} \rangle| r^2 \sin \theta d\theta d\phi \\ &= \int \int \frac{c}{4\pi} \left( \frac{kI\Delta l}{c} \right)^2 \sin^3 \theta d\theta d\phi \\ &= \frac{c}{2} \left( \frac{kI\Delta l}{c} \right)^2 \int \sin^3 \theta d\theta \\ &= \frac{2c}{3} \left( \frac{kI\Delta l}{c} \right)^2 \quad \left( \because \int \sin^3 \theta d\theta = \frac{4}{3} \right) \end{aligned}$$

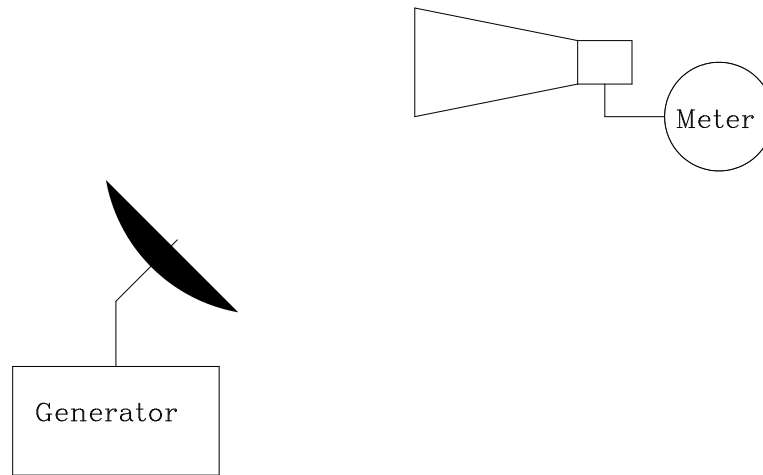
Writing as a ohmic losses of a resistor,

$$P = \frac{1}{2} RI^2$$

Then, the radiation impedance,  $R_s$  (s/cm) of the Hertz dipole is

$$R_s = \frac{2P}{I^2} = \frac{4c}{3} \left( \frac{k\Delta l}{c} \right)^2$$

# Reciprocity Theorem



A transmitting antenna powered by the generator G and a receiving antenna that induces a certain current measured by the meter M. Assume no ohmic loss. Then the current measurement on M remains the same even if we exchange generator G and meter M. Thus it does not matter which antenna is transmitting and which is receiving.

⇒ A transmitting device (e.g., Hertz Dipole) is a receiving device as well and the receiving power pattern is the same as the transmitting power pattern.

# Antenna: Power Pattern

Because of reciprocity, we can obtain the receiving power pattern of an antenna by studying the transmitting pattern. In general, the power pattern is a function of  $\theta$  and  $\phi$ :

$$P(\theta, \phi) = r^2 |\langle \mathbf{S} \rangle|$$

with a unit of (erg/s/Hz/sr) with the total radiated power (erg/s/Hz) at given  $\nu$  given by

$$W_\nu = \iint P(\theta, \phi) d\Omega$$

E.g., for a simple case like Hertz dipole, then

$$P(\theta, \phi) = r^2 |\langle \mathbf{S} \rangle| = S_0 \sin^2 \theta$$

But we usually measure the **normalized power pattern (unitless)** instead, defined as

$$P_n(\theta, \phi) = \frac{P(\theta, \phi)}{P_{\max}} \leq 1$$

# Antenna: directive gain

Define a mean power per solid angle as

$$P = \frac{\iint P(\theta, \phi) d\Omega}{4\pi}$$

then the total radiated power at given  $\nu$  is given by

$$W_\nu = 4\pi P$$

Define also a directive gain (unitless) as

$$G(\theta, \phi) = \frac{P(\theta, \phi)}{P}$$

Then we have

$$G(\theta, \phi) = \frac{4\pi}{\iint P(\theta, \phi) d\Omega} P(\theta, \phi) = \frac{4\pi}{\iint P_n(\theta, \phi) d\Omega} P_n(\theta, \phi)$$

and

$$\iint G(\theta, \phi) d\Omega = 4\pi$$

For an isotropic transmitter,  $G(\theta, \phi) = 1$ , and  $P(\theta, \phi) = P$ .

# Example: Hertz dipole

We have

$$P(\theta, \phi) = S_0 \sin^2 \theta$$

so that

$$P_n(\theta, \phi) = \sin^2 \theta$$

Since

$$\int_0^{2\pi} \int_0^\pi P_n(\theta, \phi) d\Omega = 2\pi \int_0^\pi \sin^3 \theta d\theta = \frac{8\pi}{3}$$

Thus,

$$G(\theta, \phi) = \frac{4\pi P_n(\theta, \phi)}{\iint P_n(\theta, \phi) d\Omega} = \frac{3}{2} \sin^2 \theta$$

Note  $\int \sin^3 x dx = \frac{1}{3} \cos x (\cos^2 x - 3)$

# Some Beam Definitions

The **beam solid angle (sr)**  $\Omega_A$  of an antenna is given by

$$\Omega_A = \iint_{4\pi} P_n(\theta, \phi) d\Omega$$

Here  $\Omega_A$  can be considered as a solid angle of an ideal antenna having  $P_n = 1$  and  $P_n = 0$  everywhere else.

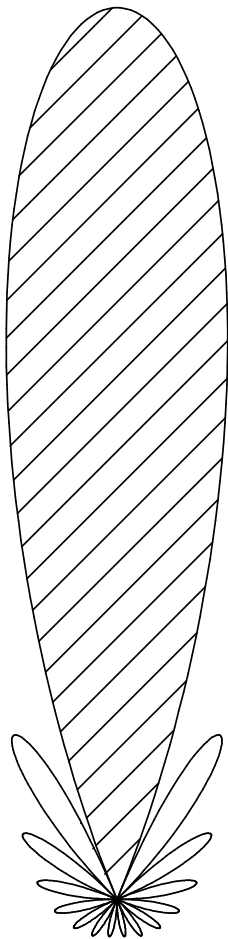
Such an antenna does not exist!!! For most antennas, the power pattern has considerably larger values for a certain range of both  $\theta$  and  $\phi$  than for the remainder of the sphere. This range is called the **main beam or main lobe** of the antenna; the remainder is called the **side lobes or back lobes**. The main beam solid angle is defined as

$$\Omega_{MB} = \iint_{MB} P_n(\theta, \phi) d\Omega$$

and the (main) beam efficiency is defined as

$$\eta_B = \frac{\Omega_{MB}}{\Omega_A}$$

indicating how much of the power pattern is in the MB.



# More on Beam pattern

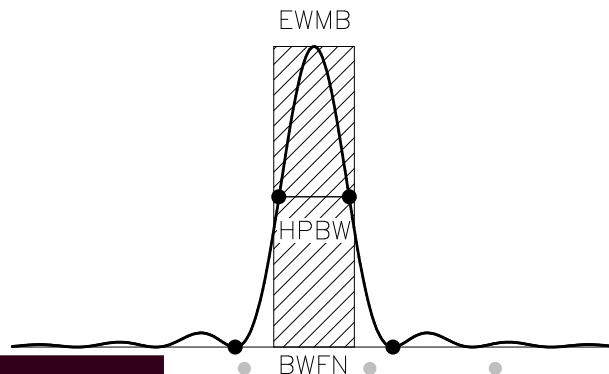
The **directive gain** is thus

$$G(\theta, \phi) = \frac{4\pi}{\Omega_A} P_n(\theta, \phi)$$

The **directivity**  $D$  defined as the max. directive gain  $G_{\max}$  is then

$$D \equiv G_{\max} = \frac{4\pi}{\Omega_A}$$

since  $P_n(\theta, \phi)$  has a max. value of 1. The directivity is a measure of how sharp the power can be focused assuming an ideal antenna. Different ways to characterize the beam pattern. HPBW: half power beam width. FWHP: full width to half power = HPBW. BWFN: beam width betw. first nulls. EWMB: equivalent width of the main beam. **Beam width is related to geometric size and  $\lambda$** , which are all we got (see next few slides).



# Effective Aperture

When a plane wave with power density  $|\langle \mathbf{S} \rangle|$  (erg/cm<sup>2</sup>/s/Hz) is intercepted by an antenna, a certain amount of power,  $P_e$  (erg/s/Hz), is extracted by the antenna. The **effective aperture** is defined as the fraction of the power being extracted

$$A_e = P_e / |\langle \mathbf{S} \rangle|$$

with a unit of cm<sup>2</sup>. Comparing this to the geometric aperture  $A_g$ , we can define an **aperture efficiency**

$$\eta_A = \frac{A_e}{A_g}$$

Effective aperture and directivity  $D$  are related by (to be proved)

$$D = \frac{4\pi A_e}{\lambda^2}$$

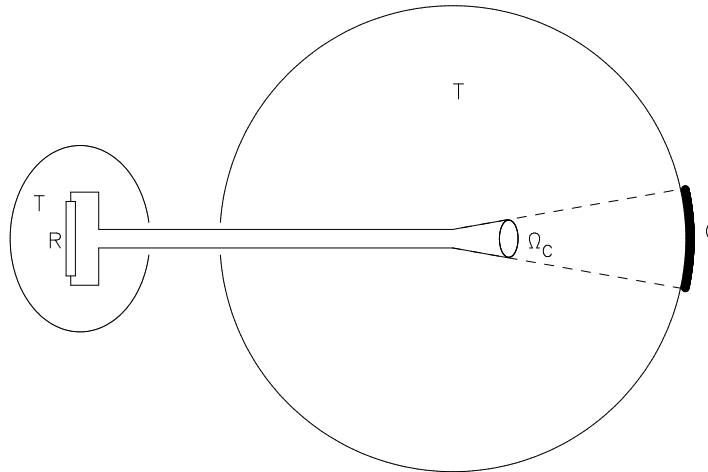
or equivalently

$$\Omega_A = \frac{\lambda^2}{A_e} \propto \left(\frac{\lambda}{d}\right)^2 \quad (\because D = G_{\max} = \frac{4\pi}{\Omega_A})$$

where  $d$  is the diameter if a circular antenna is assumed.

# Effective Aperture and Directivity

Their relationship can be derived with thermodynamic considerations:



The antenna has a directivity  $D$ , effective aperture  $A_e$  and receiver bandwidth  $\Delta\nu$ . With freq in Rayleigh-Jeans limit, the surface  $C$  radiates with the intensity

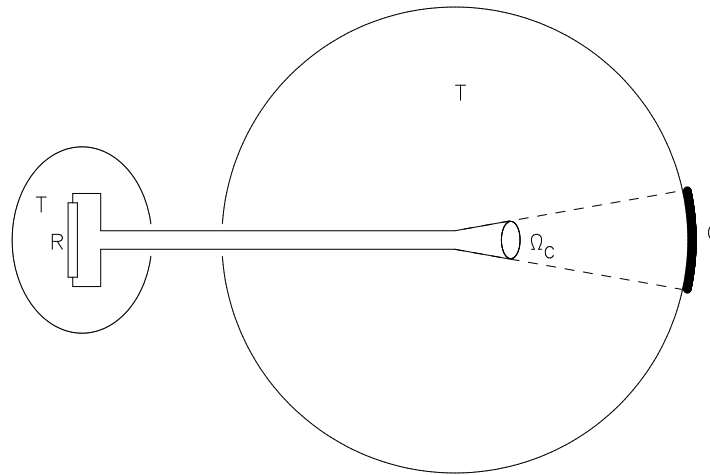
$$I = I_\nu \Delta\nu = \frac{2kT}{\lambda^2} \Delta\nu \quad (\text{erg/s/cm}^2/\text{sr})$$

The antenna collects a total power of (only one polarization component is recorded)

$$W = \frac{1}{2} A_e I \Omega_C = A_e \frac{kT}{\lambda^2} \Delta\nu \Omega_C \quad (\text{erg/s})$$

**Note:**  $\Omega_C$  is the angle subtended by the surface  $C$ .

# Effective Aperture and Directivity



The power produced by the resistor  $R$  according to the Nyquist theorem is

$$W_R = kT\Delta\nu \quad (\text{erg/s})$$

This power is radiated by the antenna with the fraction intercepted by the surface  $C$  being

$$f = \frac{\Omega_C}{\Omega_A} = D \frac{\Omega_C}{4\pi}$$

and the power intercepted by the surface  $C$  is  $W_R f$ .

# Effective Aperture and Directivity

In thermodynamic eqm, the antenna must reradiate the same amount of power into the surface C as it receives from the surface C, and thus we have  $W = W_R f$ , i.e.,

$$A_e \frac{kT}{\lambda^2} \Delta\nu \Omega_C = kT \Delta\nu D \frac{\Omega_C}{4\pi}$$

Thus,

$$D = \frac{4\pi A_e}{\lambda^2}$$

Since the relation relates quantities which do not involve thermodynamics, it should always be valid.

# Antenna Temperature

In radio astronomy, due to finite size of the beam, what we measure is **the convolution of the sky brightness with the beam**:

$$T_A(\theta_0, \phi_0) = \frac{\int T_b(\theta, \phi) P_n(\theta - \theta_0, \phi - \phi_0) \sin \theta d\theta d\phi}{\int P_n(\theta, \phi) d\Omega}$$

The brightness temperature  $T_b(\theta, \phi)$  corresponds to the thermodynamic temperature for the radiating material only for thermal radiation in the RJ limit from an optically thick source.

If the source is extended with a constant  $T_b(\theta, \phi)$ , then  $T_A = T_b$ .

